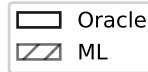
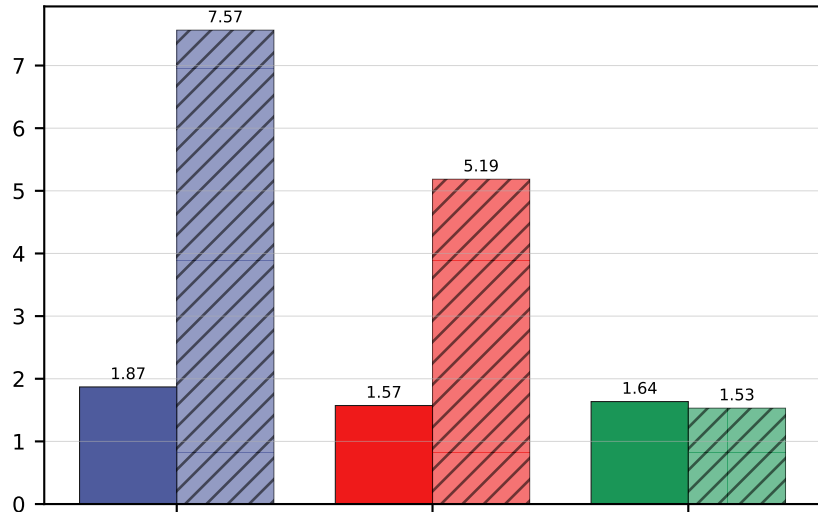


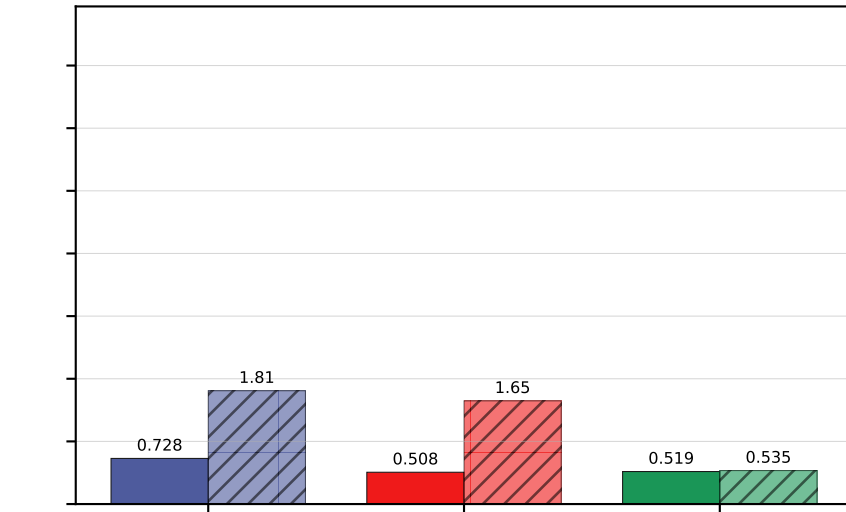
Empirical variance of ψ_{05} at the hard cell — Oracle vs ML nuisances



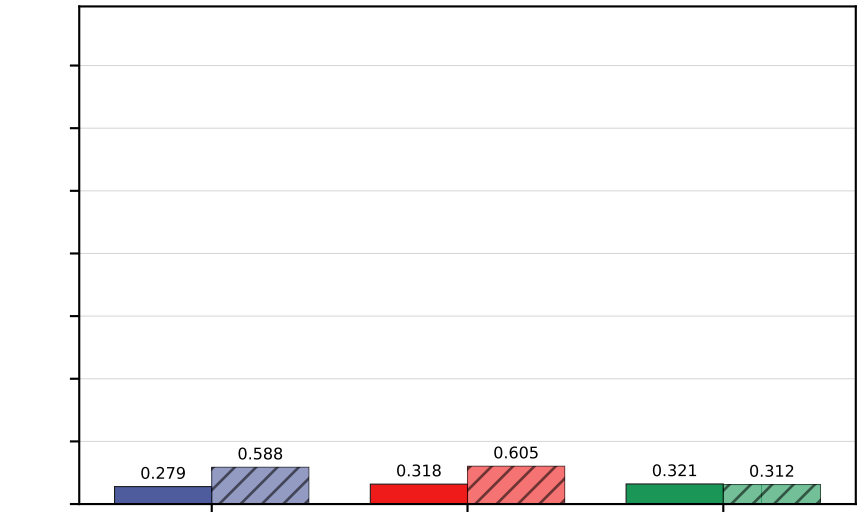
n=72,000



n=144,000

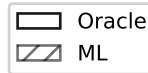


n=288,000

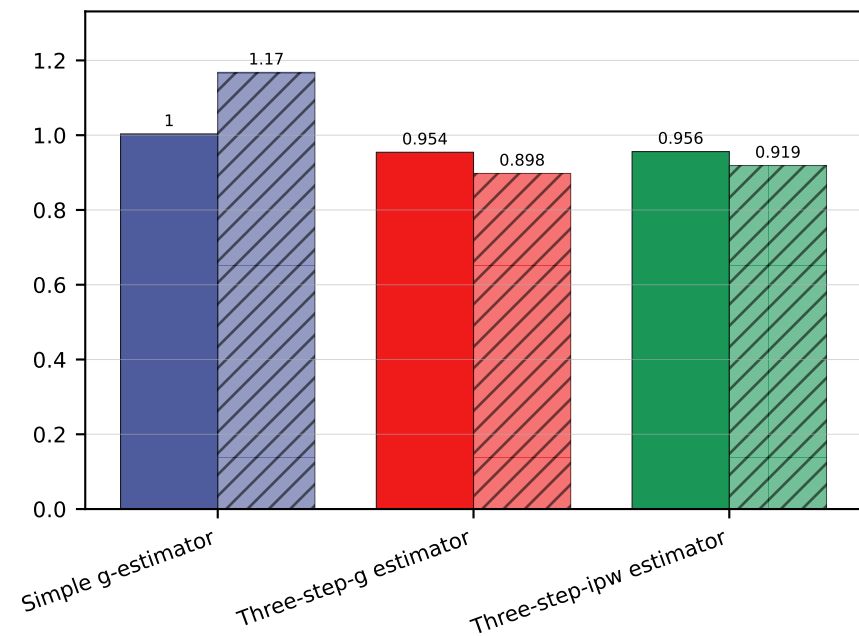


Simple g-estimator Three-step-g estimator Three-step-ipw estimator

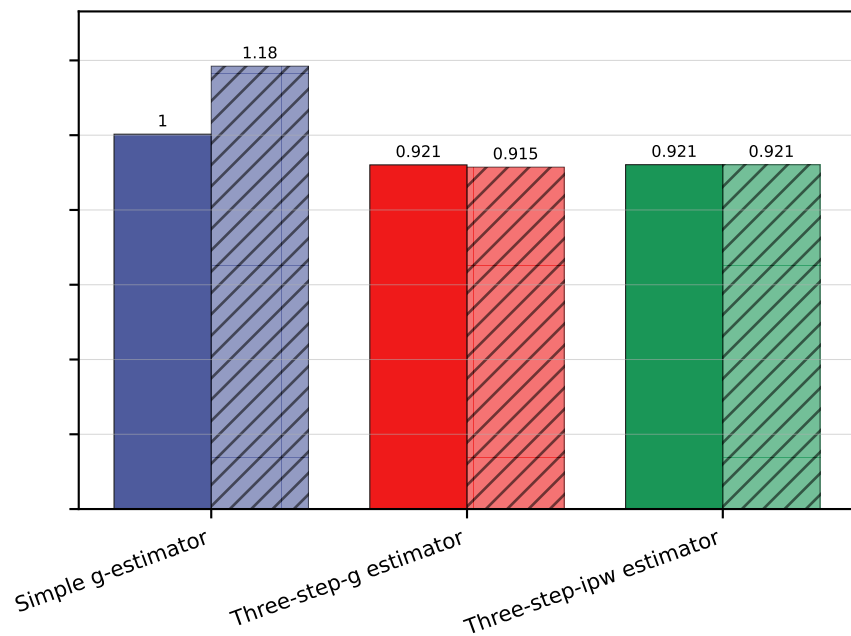
Empirical median of ψ_{05} at the hard cell — Oracle vs ML nuisances



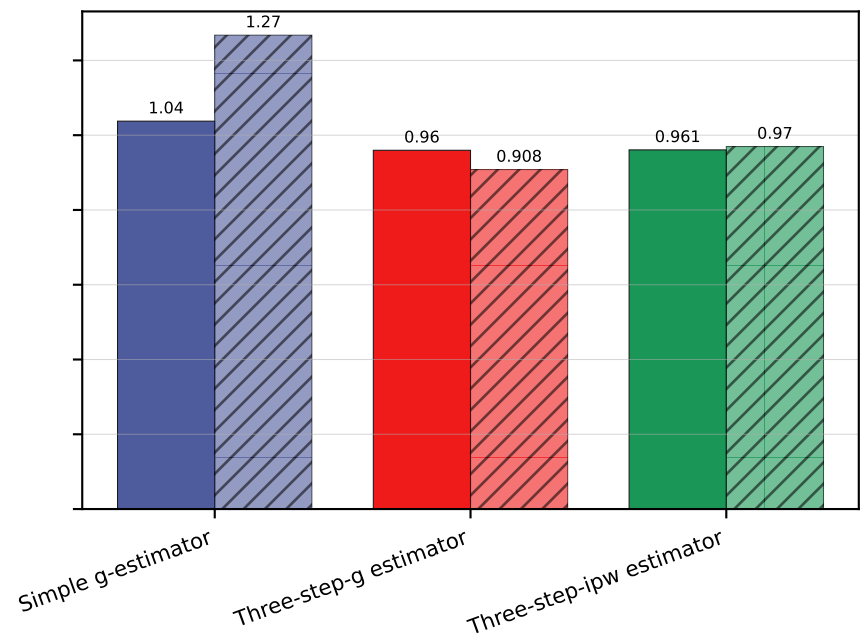
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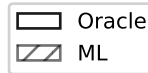
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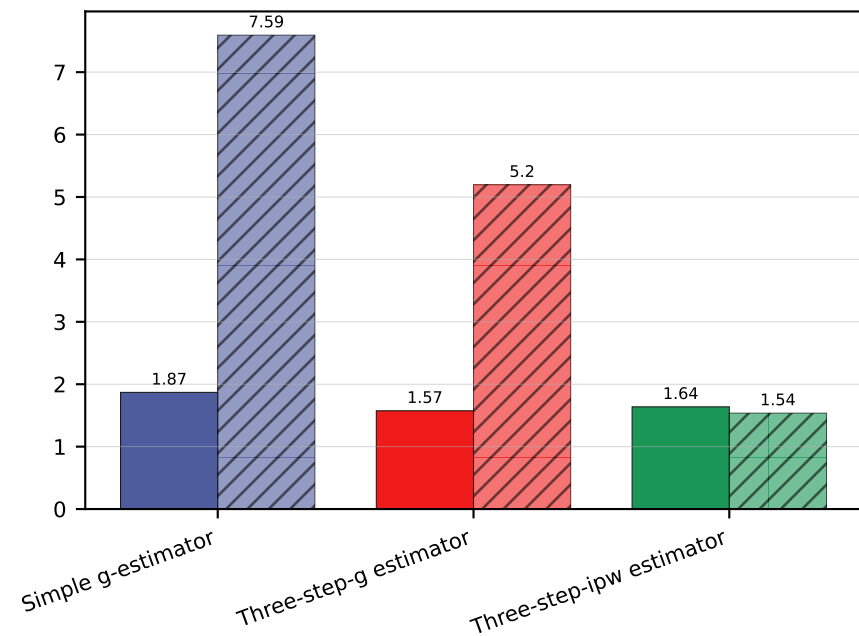
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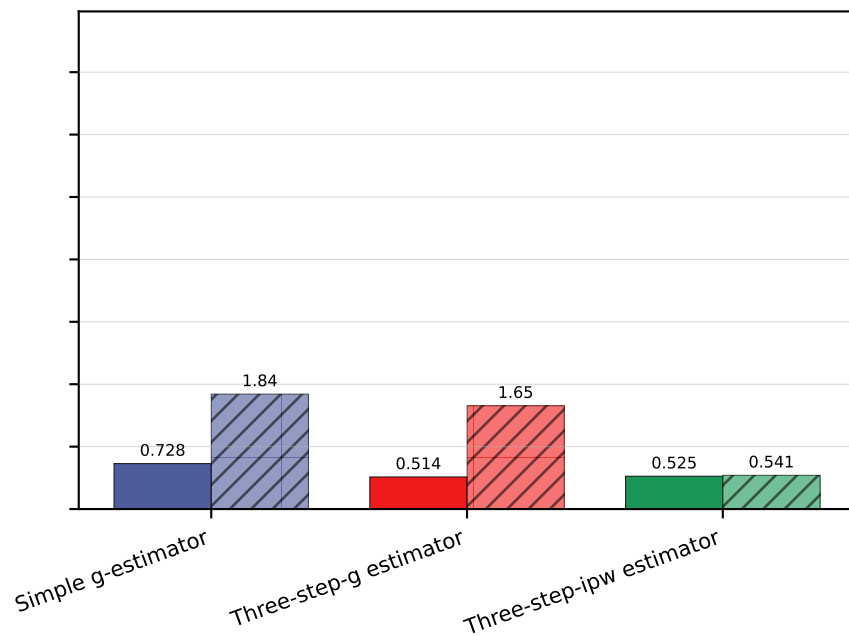
Empirical MSE of ψ_{05} at the hard cell — Oracle vs ML nuisances



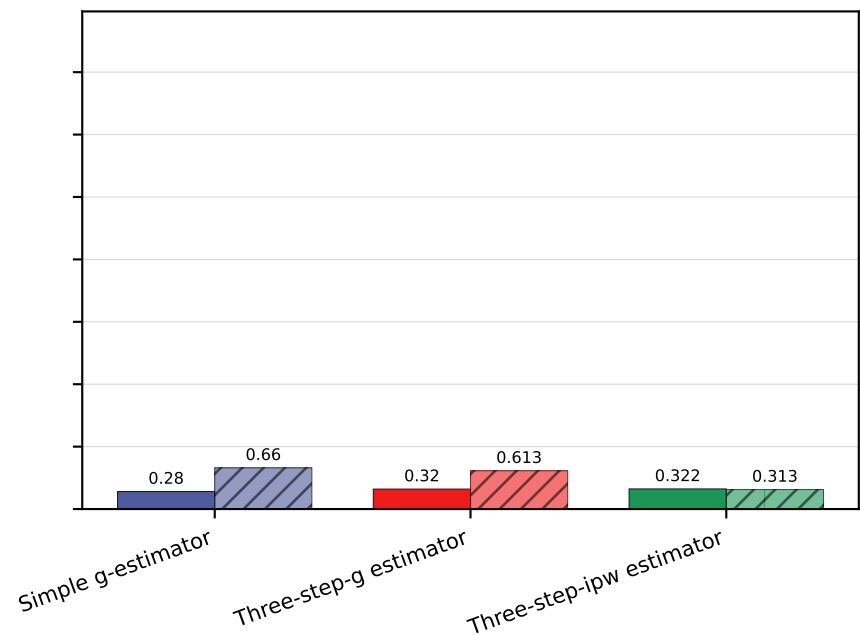
n=72,000



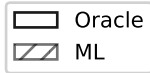
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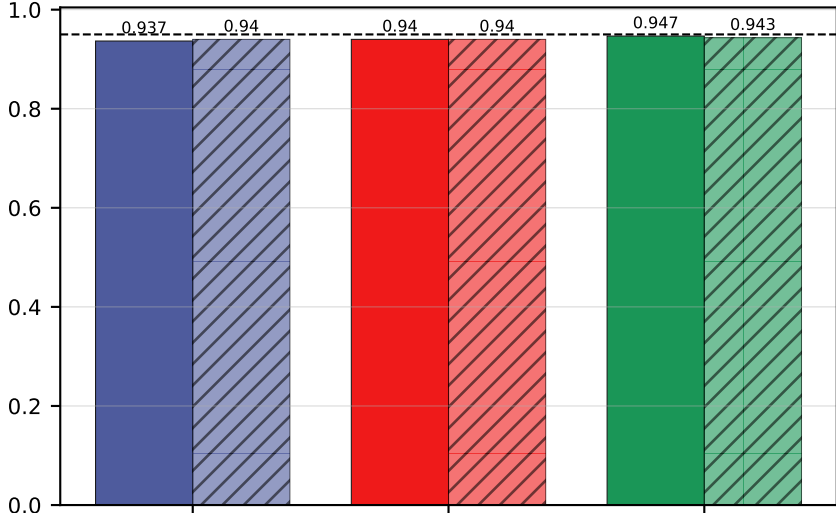
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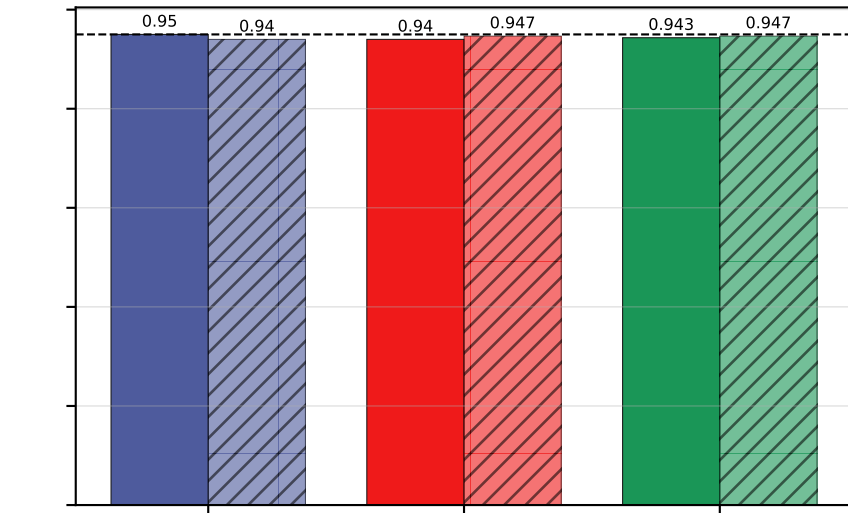
Empirical coverage of ψ_{05} at the hard cell — Oracle vs ML nuisances



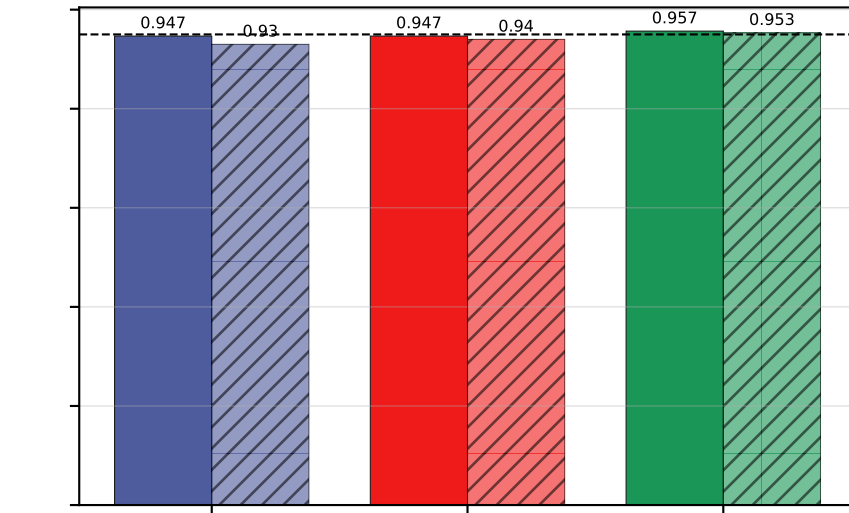
n=72,000



n=144,000



n=288,000



Simple g-estimator Three-step-g estimator Three-step-ipw estimator Simple g-estimator Three-step-g estimator Three-step-ipw estimator Simple g-estimator Three-step-g estimator Three-step-ipw estimator

Hard cell: $p_l=0.2$, $p_h=0.3$ (the worst cell in the production grid -- only $\sim 0.81\%$ of subjects follow a constant-treatment path). $n=72000$'s numbers are the hard-cell row of the full 6-cell production grid; $n=144000$ and $n=288000$ rerun only this cell, at 2x and 4x n . All runs post-ee_three_step_ipw-fix, 300 iterations, seed 3411. See CLAUDE.md "Status: $n=144000$ hard-cell check".